Amogh Ukkadgatri

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Electronics & Communication Engineer · Embedded Systems · RF & Signal Processing

SUMMARY

Electronics and communication engineer in defence R&D, on the guidance package for a small air-defence interceptor end to end: radar link-budget and detection analysis, antenna and array design with full-wave EM simulation, digital back-end architecture, and multi-board flight electronics through to fabrication-ready hardware and its measurement plan, including 24 GHz RF and substrate characterization. Background in acoustic counter-UAS — mic arrays, embedded firmware and the test rigs that prove them. Suited to regulated, high-reliability defence and aerospace work.

EXPERIENCE

R&D Intern — SSS Defence

Sep 2025 – Present

Bengaluru, India (on-site) · DeepTech R&D — Air-defence interceptor guidance package, acoustic C-UAS

- Recruited into the acoustic counter-UAS program on the strength of a final-year sound source localization project — brought in to take that array and DOA work into a fielded prototype, owning the acoustic sensing hardware from array build through measurement.
- Built the microphone arrays for the first acoustic C-UAS prototype on Raspberry Pi 4 — array geometry, sensor selection and capture across multiple communication protocols — and adapted a triangulation-based localization approach into the pipeline so the team's classical and physics-based algorithms could run on live multi-channel data.
- Built and validated the test rigs that qualify those mic arrays in hardware (see Projects): repeatable capture, channel-alignment and bearing-accuracy checks against known source positions, with DHCP, static IP and subnet masking across prototypes for reliable Ethernet transfer.
- 24 GHz RF & substrate characterization — built and validated a dedicated PCB test coupon featuring a 24 GHz patch antenna to empirically extract and measure the complex dielectric properties (ε_r and loss tangent) of high-frequency substrate materials (Rogers RO4003C) under real-world fabrication conditions.

SELECTED PROJECTS

3D Sound Source Localization using a Mic Array — Sep 2025 – Mar 2026

Real-time audio sensing system that detects sound sources and resolves their direction in three dimensions using classical DOA methods over a custom-built microphone array — array geometry and calibration, multi-channel synchronous capture and a processing chain that holds bearing accuracy on live signals in noisy rooms. The work this internship was offered on.

Real-time 3D Radar Tracking & Classification System — Sep – Nov 2024

Radar system that tracks moving objects in three dimensions and classifies them by their distinct return signatures — range–Doppler processing, track association across frames and a feature-based classifier separating object types in real time.

Theoros — Encrypted GPS Logger with LoRa Telemetry — Oct 2026

Designed a USB-powered ESP32-S3 GPS tracker end to end: a 4-layer KiCad board built for machine assembly, with a u-blox SAM-M10Q GNSS module, RFM95W 866 MHz LoRa radio, microSD and OLED, and impedance-checked RF and USB traces. Wrote the tracker and receiver firmware: LoRa telemetry and SD logs encrypted and authenticated with AES-128-CCM, plus WiFi NTP time and OTA. Ready for fabrication.

Mic-Array Validation Test Rig — ESP32 I2S PDM / TDM — 2025 – 26

Built a complete test rig single-handedly to validate custom mic arrays over ESP32 I2S PDM capture, solving the classic ESP32's single-RX-channel limit by moving to TDM.

Also: smart alcohol detection & engine immobilization — breath-alcohol sensor wired into a vehicle ignition lock (2023) · analog battery-level indicator, voltage-divider based (2024) · centroid finder in Python (2023). Full write-ups at amoghukkadgatri.vercel.app.

SKILLS

Languages & tools	C, Python, MATLAB, Verilog (basics), KiCad, Audacity, Linux
Hardware & embedded	ESP32 / ESP32-S3, Raspberry Pi, Arduino, 8051, I2S / PDM / TDM audio, LoRa, GNSS, 4-layer PCB design, sensors, op-amps, MOSFETs, transistors, soldering and firmware flashing
Systems & RF	Link-budget & detection analysis, antenna/array design, 24 GHz patch antenna & test-coupon design, substrate characterization (ε_r, loss tangent), full-wave EM simulation, digital back-end architecture, multi-board layout, signal processing
Networking	DHCP, static IP, subnet masking, Ethernet data transfer

EDUCATION

B.E., Electronics & Communication — KLE Institute of Technology · **CGPA 8.84/10** — 2022 – 2026

Senior Secondary (XII), Science — Chetan PU College, Karnataka State Board · **81.00%** — 2022

Secondary (X) — Oriental Public School, State Board · **96.16%** — 2020

LEADERSHIP & ACTIVITIES

Chair, KLEIT–IEEE MTT-S Chapter (2024–25) · Vice-Chair, KLEIT–IEEE Student Branch (2023–24) · Coordinator, Technical & Cultural Fest · Class Representative (2023–24) · University Cultural Group (Theatre)